

Justin Shiu

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EDUCATION

University of California, San Diego

La Jolla, CA

Bachelor of Science (BS), Data Science – GPA: 3.96/4.0

Expected Graduation: Jun 2027

- **Relevant Coursework:** Data Structures and Algorithms, Machine Learning, Database Systems, Software Construction, Probability & Statistics, Distributed & Scalable Systems, Optimization Methods
- **Honors:** Chancellor's Honor List (2023–2025), Dean's Honor List (All Quarters)
- **Certification:** Google Data Analytics Professional Certificate (Coursera), 2024

EXPERIENCE

Software Engineer Intern

Jun 2026 – Aug 2026

Visa Inc.

Foster City, CA

- Built and deployed a team-wide, full-stack SRE platform for Visa Intelligent Commerce (VIC) with TypeScript, Node.js, React, and Electron; supported headless execution and operational dashboards.
- Built interactive React/React Flow dashboards to visualize API latency, throughput, and transaction flows across payment services.
- Enhanced an HTTP-based MCP gateway to accelerate ClickHouse telemetry ingestion and optimize parsing before local SQLite storage, synchronizing hundreds of thousands of logs daily across 4 Visa products and 10+ containers/product.
- Tuned anomaly-detection rules across HTTP 4xx/5xx errors, latency, throughput, and TPS to improve alert quality and reduce noisy detections.
- Orchestrated LLM-powered MCP/RAG agents to correlate anomalies with recent code changes and PRs, identify root causes, recommend remediation, and flag insufficient RCA evidence.
- Generated complete root-cause analysis (RCA) reports in under one minute after developers opened anomaly notifications, reducing manual codebase investigation and SRE operational toil.

Data Science & Machine Learning Developer

Apr 2025 – Present

Association for Computing Machinery (ACM)

La Jolla, CA

- Built an item-based movie recommender in Python using 100,836 MovieLens ratings across 9,742 movies.
- Constructed user-item ratings and item-item cosine-similarity matrices for collaborative filtering.
- Cleaned and merged ratings, movie, tag, and link datasets with Pandas; standardized timestamps and genre labels for analysis.
- Developed top-N ranking logic that excluded previously rated movies and weighted item-similarity scores by users' 4+ star preferences.
- Evaluated recommendations on held-out ratings using Precision@10, Recall@10, and NDCG to assess top-K ranking quality.
- Analyzed rating distributions and low-review titles to identify data-sparsity and low-popularity limitations.

Youth Coding Instructor

Jun 2023 – Sep 2023

Each1Teach1

San Francisco, CA

- Mentored 40+ youth aged 12-16 in Scratch, Python, and JavaScript resulting in 50+ projects and 15 showcased.
- Designed weekly lesson plans aligned to rotating topics, ensuring structured progression in coding fundamentals and project development.
- Refined curriculum with 5 instructors, saving 15 hrs/week by consolidating materials and creating shared resources.

PROJECTS

Airport Database Management System | SQL, PostgreSQL, Java

- Designed a normalized PostgreSQL schema from an ER diagram modeling 7+ entities and 15+ relationships across flights, airports, customers, pilots, maintenance, and management workflows.
- Implemented role-based access control (RBAC) for management, maintenance, customers, and pilots to enforce secure, permission-based database operations.
- Built Java database access logic for relational queries and updates, preserving data integrity with constraints and primary/foreign keys while optimizing SQL queries/indexes to improve performance by 40%.
- Validated transactional workflows for flight scheduling, customer bookings, pilot assignments, and maintenance records to ensure consistent airport operations data.

F1 Driver Performance Prediction Model | Python, Scikit-learn, Pandas, Matplotlib

- Processed 6 seasons (2019–2024) of Formula 1 data and performed exploratory/correlation analysis on driver performance.
- Applied K-Means with elbow-method selection to segment drivers into 3 performance tiers.
- Collaborated on a KNN regressor using a 70/30 split and 6 features; chose $k = 3$ with test R^2 of 0.63.

Imposter Syndrome Data Science Project | Python, Pandas, Matplotlib, Statistics

- Analyzed imposter syndrome survey data from 4 CS classes (200+ responses), cleaning and transforming data with Pandas.
- Visualized patterns with Matplotlib/Seaborn and validated hypotheses using Chi-square tests and Pearson correlation.

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, JavaScript, C++, SQL, R, HTML/CSS

Frontend, Backend & Tools: React, React Flow, Electron, Node.js, FastAPI, Git, Docker, AWS, Beautiful Soup

Data Engineering & Databases: Apache Airflow, Apache Spark, Dask, ClickHouse, PostgreSQL, MySQL, SQLite

Machine Learning & Agent Systems: PyTorch, Scikit-learn, Pandas, NumPy, Statsmodels, SciPy, Seaborn, Matplotlib, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG)